

Can a Synapse Remember?

An interactive explainer for DataForge 2026

The question

AI systems usually separate slow learning from fast use: model parameters are trained, then the trained system processes new inputs. But biological brains also change their connections as activity occurs. That raises a useful computational question: can a rapidly changing synaptic state act as a short-term memory? Our explainer turns that question into a small experiment that a learner can manipulate directly.

The one-sentence claim

Recent activity can temporarily change synaptic strength, allowing a network to store short-term associations; decay and competing writes can make those memories interfere.

The experiment

The artifact contains four cue units (A–D) and four target units (X–W). A learner writes an association such as A→X. The current synaptic matrix stores the strength of every cue-target connection. Before each write, existing fast memory is multiplied by a retention factor. The selected association then receives a Hebbian-style increment. The learner can query the matrix at any time. The system returns the strongest target and places it beside the intended ground truth.

Why the failure case matters

A memory mechanism is more informative when it can fail. If A→X is written and then A→Y is repeatedly written, the Y synapse can become stronger than X. If enough time steps pass, both traces decay. These failures expose two properties of fast memory: it is plastic enough to adapt quickly, and therefore vulnerable to interference and forgetting. The artifact deliberately makes those effects visible rather than hiding them behind a final accuracy number.

The mathematical toy model

The implementation uses the compact update $w(t+1) = r \cdot w(t) + \eta \cdot \text{pre}(t) \cdot \text{post}(t)$. Here r is a retention factor between zero and one, η is the write strength, and pre/post are one-hot activity indicators for the selected cue and target. This is a teaching simplification, not a claim that the full biological synapse or BDH uses exactly this four-by-four rule.

Connection to Dragon Hatchling

The connection to BDH is substantive. Kosowski et al.'s primary BDH paper describes a brain-inspired post-Transformer architecture whose working memory during inference relies on synaptic plasticity with Hebbian learning using spiking neurons. The paper also reports that individual synapses strengthen when BDH processes specific concepts. Thus the concept shown in the toy system corresponds to a named architectural mechanism in BDH. The correct scientific statement is not “our demo is BDH”; it is “our demo isolates and makes manipulable one mechanism that the BDH paper uses in its architecture.”

Current research context

Short-term plasticity has also appeared directly in machine learning. Rodriguez, Guo and Moraitis (ICML 2022) proposed Short-Term Plasticity Neurons whose synapses have a state propagated through time, framing the mechanism as a way to learn and forget in the short term. Brito and Gerstner (2024) developed a computational theory of synaptic plasticity that addresses how learning rules can extract useful structure from correlated inputs. These works show that plasticity is an active computational research area, while the BDH paper provides a recent architecture-level case study.

What the learner should leave with

The learner should be able to say: a synapse can act like a small, fast-changing memory variable; recent co-activity can strengthen an association; decay removes old information; competing writes can cause interference. They should also be able to distinguish this mechanism from permanent parameter learning and from the complete BDH system.

Limitations

The toy model is intentionally tiny. It does not reproduce BDH's neuron dynamics, graph structure, sparse positive activations, training regime or benchmark results. Its purpose is pedagogical: expose the variables that matter and let the learner challenge the central claim in under a minute. Published BDH results are therefore kept separate from the live toy computation and labeled as published evidence rather than reproduced evidence.

References

Kosowski et al. (2025). The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain. <https://arxiv.org/abs/2509.26507>

Rodriguez, Guo & Moraitis (2022). Short-Term Plasticity Neurons Learning to Learn and Forget. <https://proceedings.mlr.press/v162/rodriguez22b.html>

Brito & Gerstner (2024). Learning what matters: Synaptic plasticity with invariance to second-order input correlations. <https://doi.org/10.1371/journal.pcbi.1011844>

Engdahl et al. (2026). BDH-CQ: In-Context Learning with Recurrent Latent Reasoning. <https://arxiv.org/abs/2608.09888>